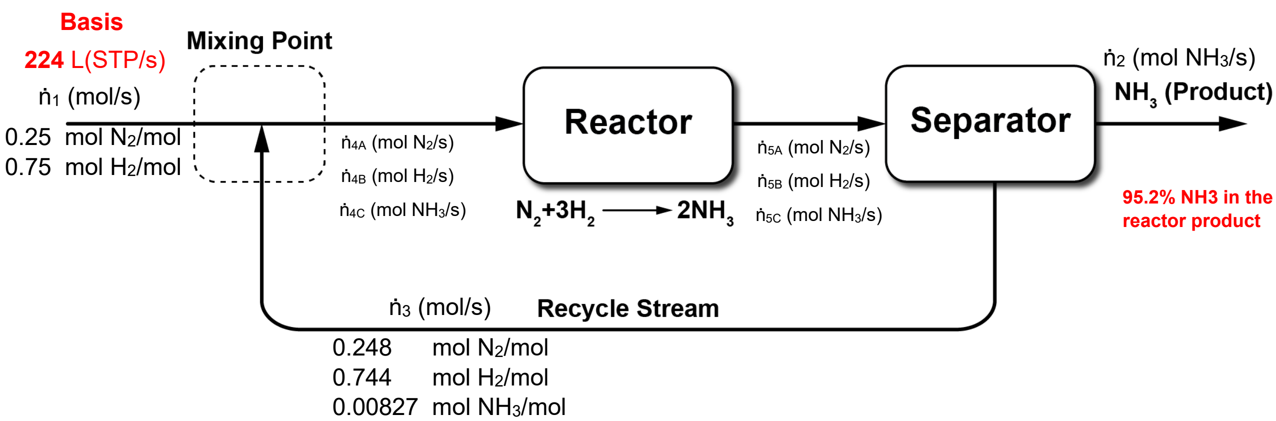
Test #2

1. **(55 pts total)**

a) **(10 pts)**



b) **(4 pts)**

$$X_{Overall} = \frac{0.25\dot{n}_1 - 0}{0.25\dot{n}_1} \times 100\% = 100\%$$

$$X_{Single-Pass} = \frac{\dot{n}_{4A} - \dot{n}_{5A}}{\dot{n}_{4A}} \times 100\%$$

c) **(4 pts)**

2 unknowns ($\dot{n}_1$, $\dot{n}_2$)
-1 independent atomic balance (N or H)
-1 ideal gas EOS

0 DOF

d) **(3 pts)**

N and **H** are not independent

H Balance**= 3 × N** Balance

e) **(4 pts)**

Ideal gas EOS : $\dot{n}_1 = \frac{224\ L(STP)}{s} \times \frac{1\ mol}{22.4\ L(STP)} = 10\ mol/s$

N Balance : $\left(10.0\frac{mol}{s}\right)\left(0.25\frac{mol\ N_2}{mol}\right) = \dot{n}_2\frac{mol\ NH_3}{s}$

$$\boldsymbol{\dot{n}_2 = 5.00\ mol\ NH_3/s}$$

f) (15 pts)

Mixing Point

4 unknowns ($\dot{n}_3$, $\dot{n}_{4A}$, $\dot{n}_{4B}$, $\dot{n}_{4C}$)
- 3 independent balances (N_2, H_2, NH_3, and tot: 3 out of 4)
1 DOF

Reactor

6 unknowns ($\dot{n}_{4A}$, $\dot{n}_{4B}$, $\dot{n}_{4C}$, $\dot{n}_{5A}$, $\dot{n}_{5B}$, $\dot{n}_{5C}$)
- 2 independent atomic balance (N and H)
4 DOF

Separator

4 unknowns ($\dot{n}_3$, $\dot{n}_{5A}$, $\dot{n}_{5B}$, $\dot{n}_{5C}$)
- 3 independent balances (N_2, H_2, NH_3, and tot: 3 out of 4)
- 1 extra info (95.2%)
0 DOF

Strategy: Analyze the separator unit first, and then the mixing point.

g) (8 pts)

Separator:

$$\begin{cases} 95.2\% \rightarrow \ \dot{n}_2 = 0.952\dot{n}_{5c} & (1) \\ N_2\ balance\colon \dot{n}_{5A} = 0.248\dot{n}_3 & (2) \\ H_2\ balance\colon \dot{n}_{5B} = 0.744\dot{n}_3 & (3) \\ NH_3\ balance\colon \dot{n}_{5C} = \dot{n}_2 + 0.008278\dot{n}_3 & (4) \end{cases}$$

h) (3 pts)

A **purge** line needs to be added to the process to avoid build-up of Argon in the system.

2. (12 pts total)

a) (3 pts)

Standard Condition: T_s=273.15 K P_s=1 atm

After Compressor: T_1=136.55 K P_1=10 atm

Ideal Gas EOS:

$$\frac{P_s \dot{V}_s}{T_s} = \frac{P_1 \dot{V}_1}{T_1} \rightarrow \dot{V}_1 = \frac{(P_s/P_1)\, \dot{V}_s}{T_s/T_1} = \mathbf{10\ m^3/min}$$

b) (7 pts)

$T_{c,O2}$=154.4 K $P_{c,O2}$=49.7 atm
$T_{c,N2}$=126.2 K $P_{c,N2}$=33.5 atm

Kay's Rule:

$$T'_c = (0.21 \times 154.6) + (0.79 \times 126.2) = 132.64\ K$$
$$P'_c = (0.21 \times 49.7) + (0.79 \times 33.5) = 36.9\ atm$$

$$T'_r = \frac{T_1}{T'_c} = \frac{136.55}{132.64} = 1.03$$
$$P'_r = \frac{P_1}{P'_c} = \frac{10}{36.9} = 0.27$$

➔ Compressibility Factor, z=0.92
(5.4-2)

$$\dot{V}_1 = \frac{znRT_1}{P_1} = z \times \frac{nRT_1}{P_1} = 0.92 \times 10\ m^3/min = \mathbf{9.2\ m^3/min}$$

c) (2 pts)

No, since the compressibility factor EOS gives 10% less flow.

OR

RT/P=1.1<5

3. **(18 pts total)**

a) **(4 pts)**

$$C_3H_8+5O_2 \rightarrow 3CO_2+4H_2O$$
$$C_3H_8+7/2O_2 \rightarrow 3CO+4H_2O$$

b) **(4 pts)**

6 unknowns ($\dot{n}_1$, $\dot{n}_2$, $\dot{n}_3$, $\dot{n}_4$, $\dot{n}_5$, $\dot{n}_6$)
-3 independent atomic balance (C, H, O)
-1 inert species balance (N_2)
-1 conversion of propane (90%)
-1 ideal gas EOS (inlet feed → $\dot{n}_1$)

0 DOF

c) **(2 pts)**

Yes, $\frac{RT}{P} = \frac{(0.08206\ L.atm/mol\ K)\ (200\ ℃+273.15)K}{1.5\ atm} = \boldsymbol{25.87 > 20}$

d) **(10 pts)**

$$\begin{cases} IG\ EOS:\ \dot{n}_1 = \dfrac{V(STP)}{\hat{V}_s} \quad or\ P_s V(STP) = \dot{n}_1 RT_s & (1) \\ C\ balance: 3\dot{n}_1 = 3\dot{n}_2 + \dot{n}_4 + 10 & (2) \\ H\ balance: 8\dot{n}_1 = 8\dot{n}_2 + 2\dot{n}_5 & (3) \\ O\ balance: 2(0.21)200 = 2\dot{n}_3 + 2\dot{n}_4 + \dot{n}_5 + 10 & (4) \\ N_2\ balance: 0.79(200) = \dot{n}_6 & (5) \\ 90\%\ conversion: \dot{n}_2 = 0.1\dot{n}_1 & (6) \end{cases}$$

4. **(10 pts total)**

a) **(5 pts)**

$$\dot{n} = \frac{\dot{V}(STP)}{\hat{V}_s} = \frac{44.8\ m^3/hr\ (STP)}{22.4\ m^3/kmol} \times \frac{1000\ mol}{1\ kmol} = \boldsymbol{2000\ mol/hr}$$

b) **(5 pts)**

$$C_2H_4+0.5O_2 \rightarrow C_2H_4O$$
$$\frac{50\ lbmol\ C_2H_4}{1} \times \frac{0.5\ mol\ O_2}{1\ mol\ \ C_2H_4} = 25\ lbmol\ O_2$$

So, O_2 is provided in excess.

$$\frac{50\ lbmol\ O_2 - 25\ lbmol\ O_2}{25\ lbmol\ O_2} \times 100\% = \boldsymbol{100\ \%\ excess\ O_2}$$